

Mshauri - Fresh VPS Installation Guide

Deployment target: mshauri.marichomedia.com | VPS: 64.247.178.45 | Ubuntu 22.04

Validated deployment notes

- Full TypeScript validation passed for all workspace packages.
- API production build passed: artifacts/api-server/dist/index.mjs.
- Frontend build passed: artifacts/mhauri-ai/dist/public.
- Deployment scripts pass shell checks and generate DATABASE_URL automatically.

The final VPS verification still must be performed on the server because SSH access is required to confirm its database, Nginx, PM2, DNS, and SSL status.

1. Before you begin

- You need DB_PASS and SESSION_SECRET for the database and secure user sessions. Generate both on the VPS in Step 5.
- OPENAI_API_KEY is optional for the website but required for Ask Mshauri AI chat.
- WhatsApp settings are optional for the website but required for the WhatsApp bot: WHATSAPP_ACCESS_TOKEN, WHATSAPP_PHONE_NUMBER_ID, and WHATSAPP_VERIFY_TOKEN.

2. Point the domain to the VPS

- In the DNS manager for marichomedia.com, create or update an A record: Type A, Name mshauri, Value 64.247.178.45, TTL 300 (or Auto).
- Wait for DNS to update before asking for the SSL certificate.

```
nslookup mshauri.marichomedia.com
# Expected IP: 64.247.178.45
```

3. Log in and download the current code

- Use the first command block for a fresh server. If /var/www/mshauri already exists, use the second block instead.

```
ssh root@64.247.178.45
cd /var/www
git clone https://github.com/seanndlovu/Mshauri-AI-Advisor.git mshauri
cd /var/www/mshauri

cd /var/www/mshauri
git pull origin main
```

4. Install server software

- This installs Node.js 24, pnpm, PM2, PostgreSQL, Nginx, and Certbot. It takes several minutes.
- If another web server such as Apache is using port 80, stop it before proceeding.

```
cd /var/www/mshauri
sudo bash deploy/1-install-vps.sh

sudo lsof -i :80
# Only if apache2 appears in the output:
sudo systemctl stop apache2
sudo systemctl disable apache2
```

5. Generate production secrets

- Copy the two outputs somewhere safe. They will be pasted into the production settings file in the next step.

```
openssl rand -hex 24 # DB_PASS
openssl rand -hex 32 # SESSION_SECRET
```

6. Create the production settings file

- Copy the template then edit it. Replace only the placeholder values.
- Do NOT add DATABASE_URL. The updated deployment scripts generate it safely and automatically from DB_PASS.
- Save Nano with Ctrl+X, then Y, then Enter.

```
cd /var/www/mshauri
cp deploy/.env.example deploy/.env.production
nano deploy/.env.production

PORT=8080
NODE_ENV=production
DOMAIN=mshauri.marichomedia.com

DB_NAME=mshauri
DB_USER=mshauri
DB_PASS=PASTE_THE_FIRST_OPENSSL_VALUE_HERE

SESSION_SECRET=PASTE_THE_SECOND_OPENSSL_VALUE_HERE

OPENAI_API_KEY=sk-PASTE_YOUR_OPENAI_KEY_HERE

WHATSAPP_ACCESS_TOKEN=PASTE_META_ACCESS_TOKEN_HERE
WHATSAPP_PHONE_NUMBER_ID=PASTE_META_PHONE_NUMBER_ID_HERE
WHATSAPP_VERIFY_TOKEN=YOUR_PRIVATE_VERIFY_TOKEN
```

7. Build and start Mshauri

- This command creates the database and database user, installs dependencies, builds the API and frontend, applies the schema, creates the Nginx site, and starts the API through PM2.
- Expected end message: Mshauri is running!

```
cd /var/www/mshauri
bash deploy/2-setup-app.sh
```

8. Enable HTTPS

- Only do this after the DNS command in Step 2 returns 64.247.178.45.
- When asked whether to redirect HTTP to HTTPS, choose option 2 (Redirect).

```
sudo certbot --nginx -d mshauri.marichomedia.com
sudo certbot renew --dry-run
```

9. Add starter communities and example posts (optional)

- Run this once if you want the live platform to begin with farming communities and sample discussions. Skip it if you want to start with an empty site.

```
cd /var/www/mshauri
pnpm --filter @workspace/scripts run seed:community
```

10. Final verification

- Open <https://mshauri.marichomedia.com> in a browser after these checks pass.
- Test registration/sign-in, community pages, and Ask Mshauri AI if an OpenAI key was added.

```
pm2 status
curl http://localhost/api/healthz
sudo nginx -t
curl -I https://mshauri.marichomedia.com
```

11. Future updates

- After new code has been pushed to GitHub, run the following. The update script pulls the new code, rebuilds it, applies database schema changes, and restarts the API.

```
cd /var/www/mshauri
bash deploy/3-update.sh
```

Troubleshooting

Problem	Check or fix
Browser shows 502 / API is down	pm2 logs mshauri-api --lines 100 pm2 restart mshauri-api --update-env
Domain does not open	nslookup mshauri.marichomedia.com # Must return 64.247.178.45
SSL certificate fails	Fix DNS first, wait for propagation, then rerun Certbot.
Nginx configuration error	sudo nginx -t sudo systemctl reload nginx
Build runs out of memory	sudo fallocate -l 2G /swapfile && sudo chmod 600 /swapfile && sudo mkswap /swapfile && sudo swapon /swapfile

GitHub source: <https://github.com/seanndlovu/Mshauri-AI-Advisor>